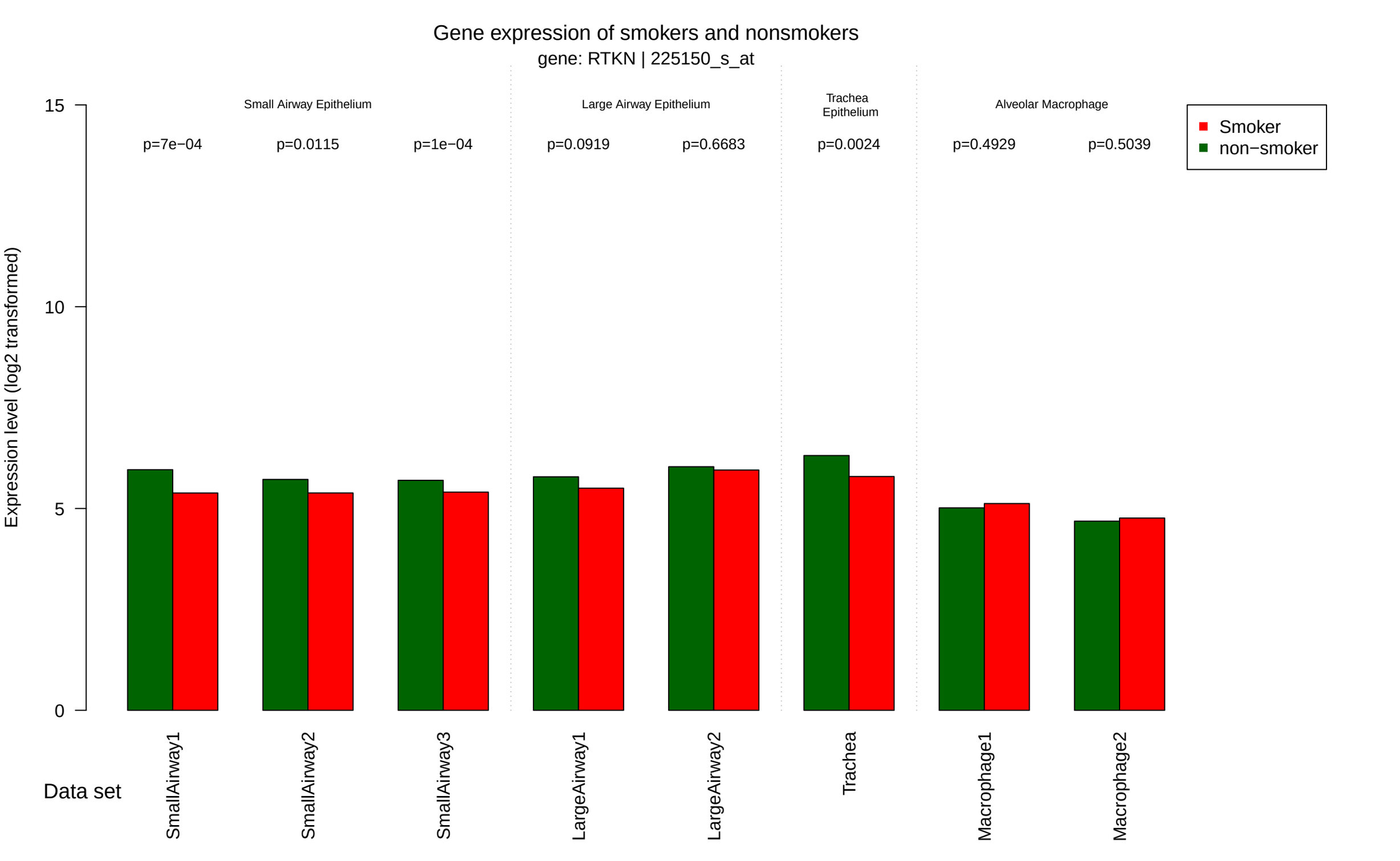
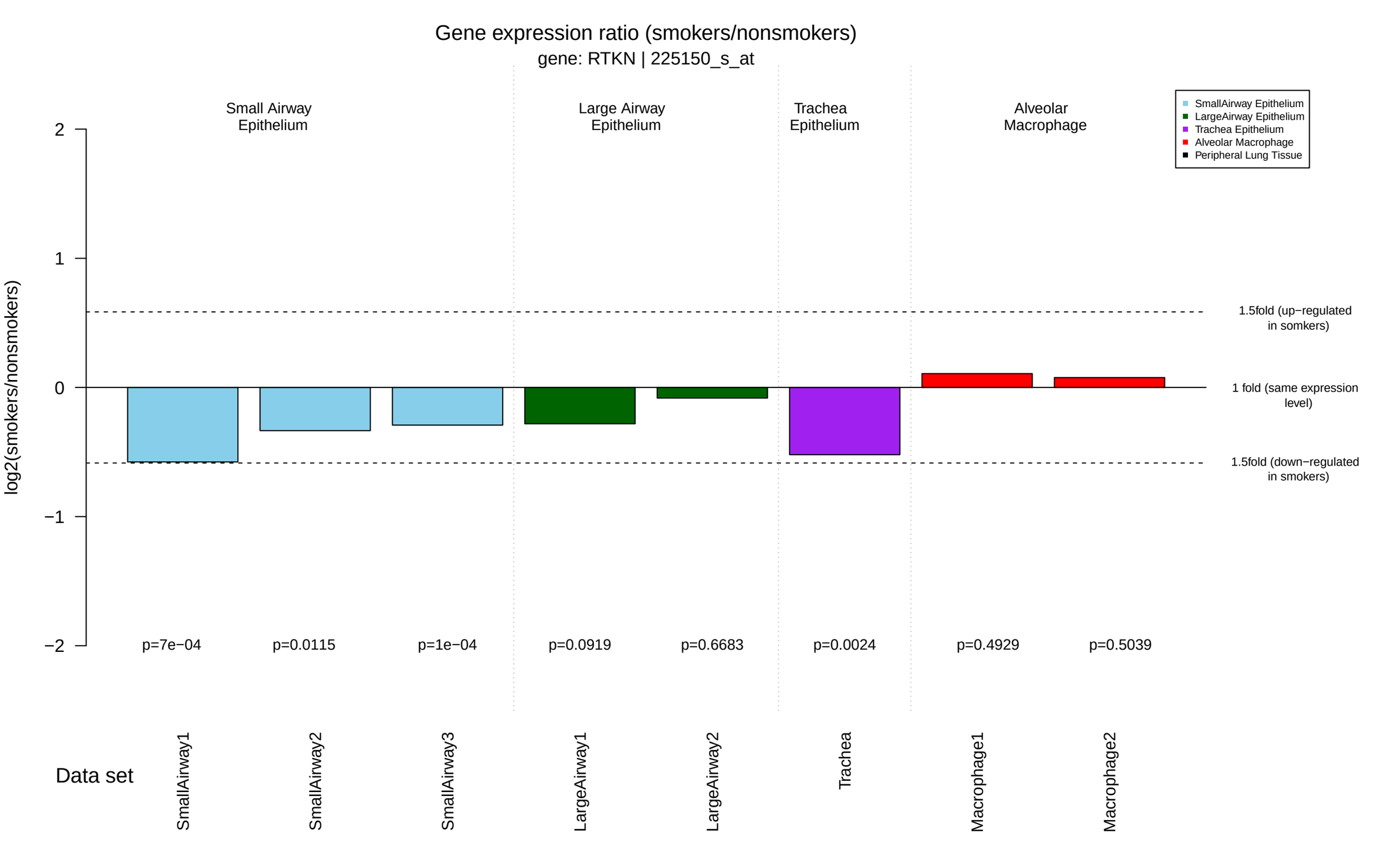
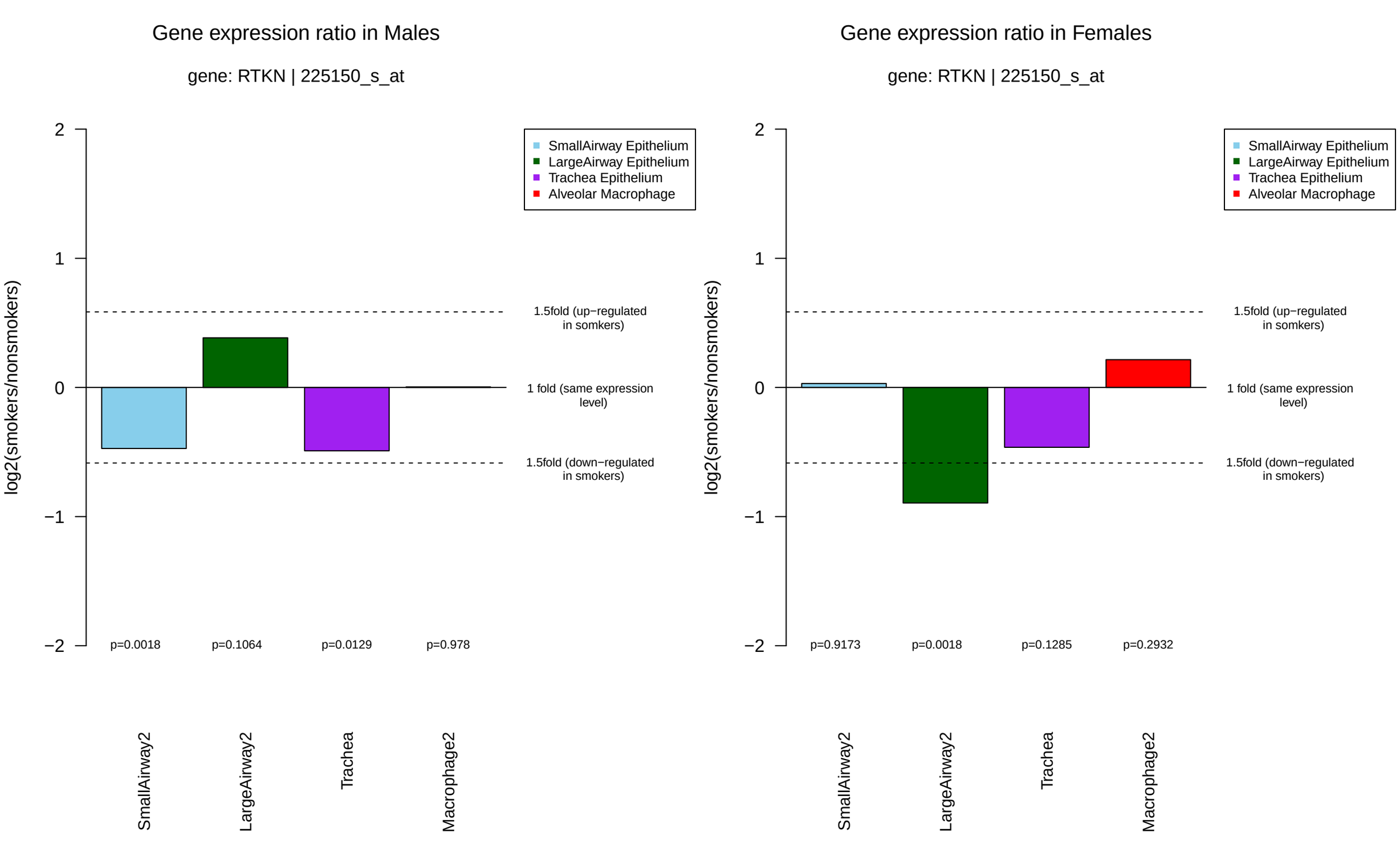
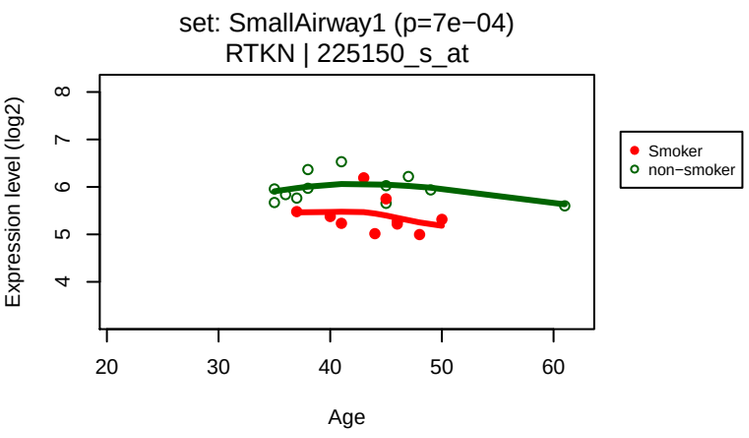
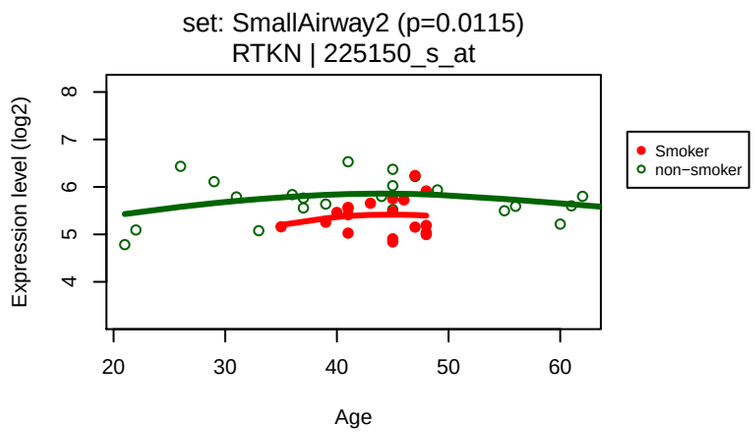
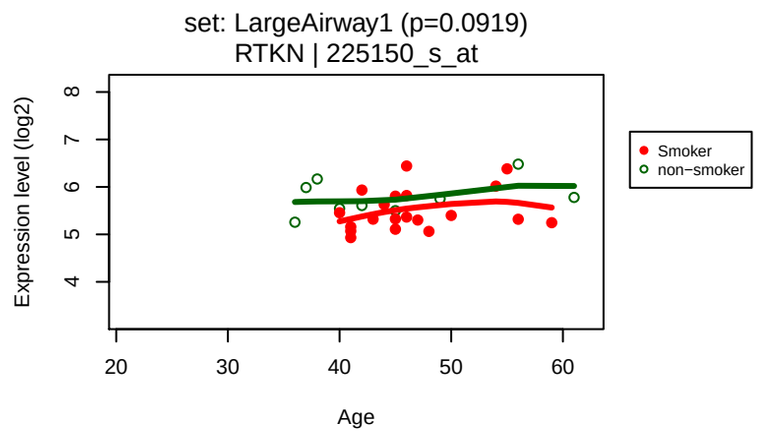

Gene expression of smokers and nonsmokers
gene: RTKN | 225150_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=7e−04
p=0.0115
p=1e−04
p=0.0919
p=0.6683
p=0.0024
p=0.4929
p=0.5039
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

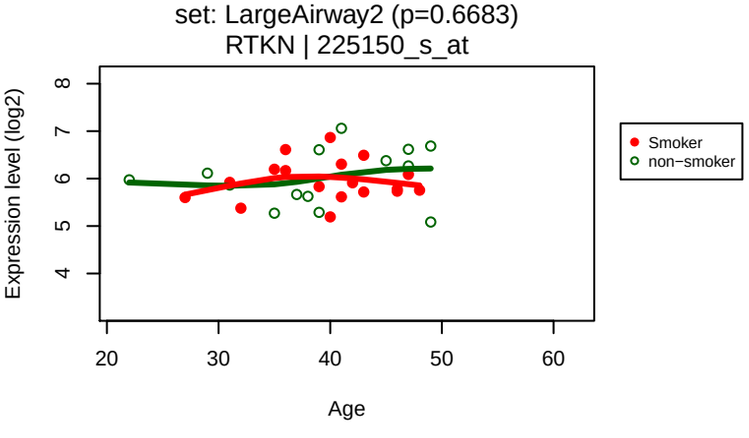

Gene expression ratio (smokers/nonsmokers)
gene: RTKN | 225150_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=7e−04
p=0.0115
p=1e−04
p=0.0919
p=0.6683
p=0.0024
p=0.4929
p=0.5039
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

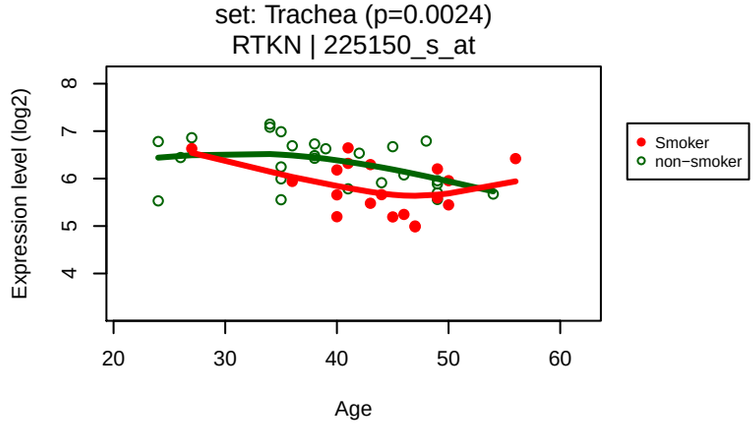

Gene expression ratio in Males
gene: RTKN | 225150_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0018
p=0.1064
p=0.0129
p=0.978
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RTKN | 225150_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9173
p=0.0018
p=0.1285
p=0.2932
SmallAirway2
LargeAirway2
Trachea
Macrophage2

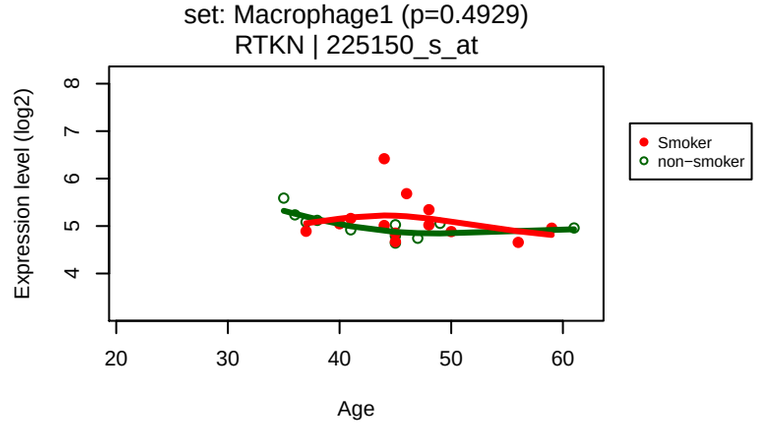

set: SmallAirway1 (p=7e−04)
RTKN | 225150_s_at
Expression level (log2)
Age
Smoker
non−smoker

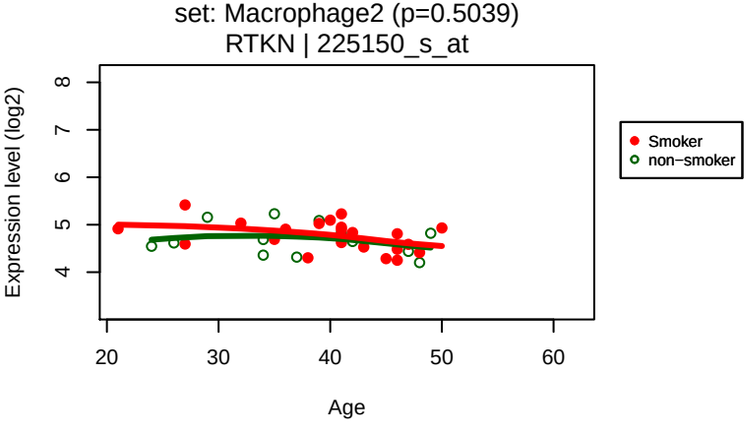

set: SmallAirway2 (p=0.0115)
RTKN | 225150_s_at
Expression level (log2)
Age
Smoker
non−smoker

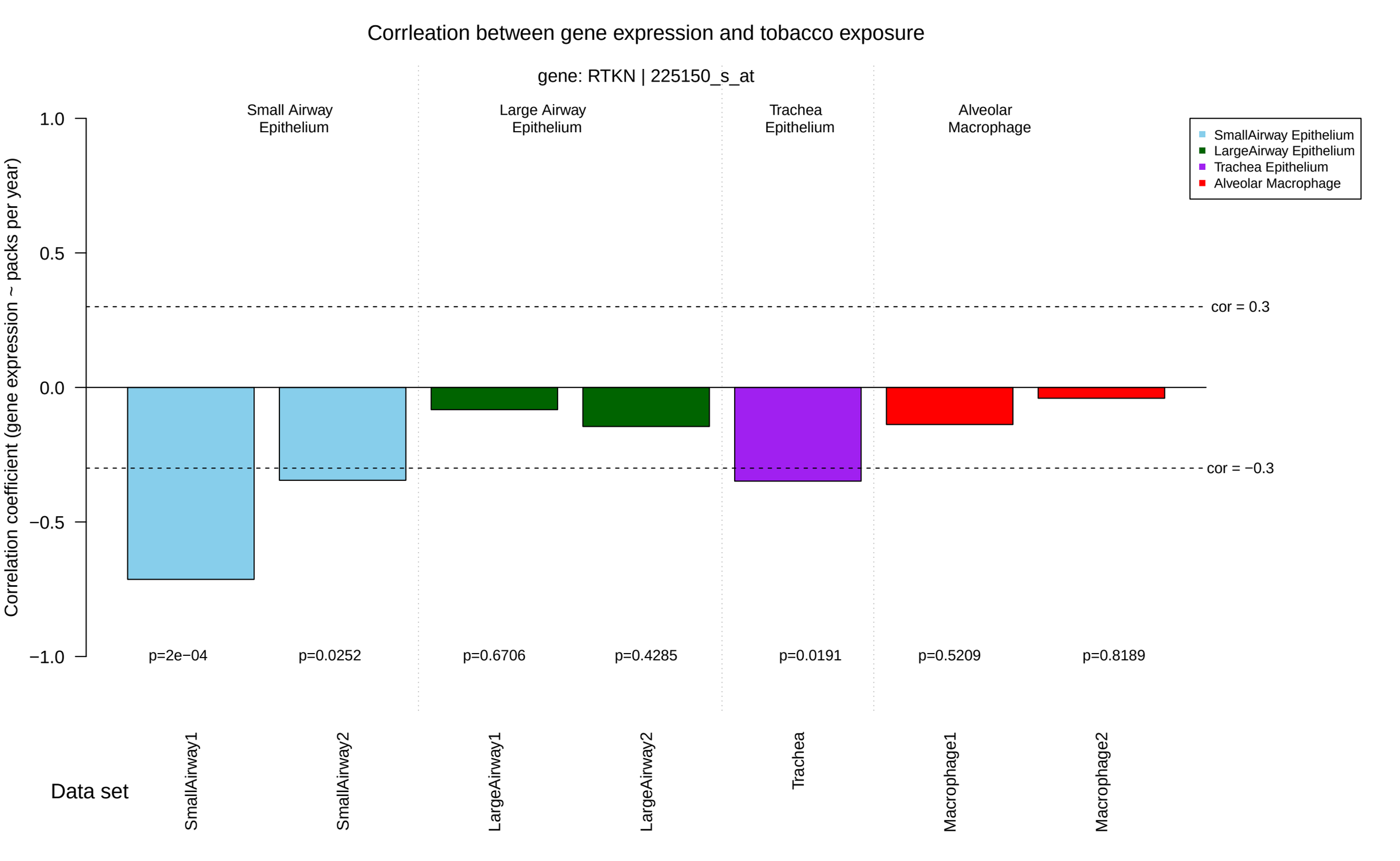

set: LargeAirway1 (p=0.0919)
RTKN | 225150_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6683)
RTKN | 225150_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0024)
RTKN | 225150_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4929)
RTKN | 225150_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5039)
RTKN | 225150_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RTKN | 225150_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=2e−04
p=0.0252
p=0.6706
p=0.4285
p=0.0191
p=0.5209
p=0.8189
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2